

Minimal-Cost Experimental Test of EVRT Signatures in a Tabletop Resonator

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This paper proposes a low-cost, tabletop experimental pathway for testing whether coherent nonequilibrium electromagnetic resonators exhibit measurable response deviations consistent with Emergent Vacuum Response Theory (EVRT). Rather than requiring cryogenic operation or superconducting cavities, the proposed experiment uses a moderate-Q resonator, calibrated ringdown measurement, and controlled parameter sweeps, including baseline and control-condition comparisons. The objective is not to demonstrate new physics, but to establish a first-pass falsification platform capable of identifying small, bounded, and reproducible deviations in ringdown time, phase lag, and linewidth behavior. A null result constrains EVRT-like response under accessible laboratory conditions, while any candidate signal would require higher-precision follow-up.

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I. INTRODUCTION

Emergent Vacuum Response Theory (EVRT) proposes that coherent nonequilibrium electromagnetic systems may exhibit small, resonance-dependent deviations from ordinary linear electromagnetic response. Prior work has emphasized conservation compatibility, Maxwellian recovery limits, and falsifiable observables such as delayed ringdown, phase lag, and linewidth variation.

The present work does not assume any specific microscopic mechanism, but instead focuses on experimentally accessible phenomenological signatures.

High-precision experimental tests often require cryogenic environments and high-Q superconducting cavities. Before such complexity is justified, a minimal tabletop experiment can serve as a screening test. The goal is not confirmation, but falsification or constraint.

This work serves as a minimal experimental extension of the EVRT framework developed in prior work, focusing specifically on low-cost falsification pathways.

Even in the absence of a positive signal, such measurements provide meaningful constraints on the parameter space of possible nonequilibrium electromagnetic response.

II. EXPERIMENTAL OBJECTIVE

The experiment tests whether a tabletop resonator exhibits deviations from standard exponential cavity decay under controlled conditions.

Primary observables:

1. ringdown deviation relative to exponential decay,
2. phase lag relative to drive,
3. linewidth variation near resonance.

A valid signal must be:

- reproducible,
- parameter-dependent,
- absent under control conditions.

III. MINIMAL APPARATUS

A practical tabletop setup includes:

- RF signal generator (10 MHz – few GHz range),
- resonant cavity or LC resonator ($Q \sim 10^3$ – 10^5),
- directional coupler,
- fast RF switch for ringdown measurement,
- oscilloscope or digitizer,
- temperature sensor,
- optional permanent magnet or coil.

The system is operated in ambient or lightly controlled temperature conditions.

IV. MEASUREMENT PROTOCOL

The experimental sequence is:

1. establish baseline resonance and Q-factor,
2. drive the resonator at resonance,
3. abruptly terminate the drive,
4. record ringdown amplitude $A(t)$,
5. repeat across multiple trials to establish statistical consistency,
6. vary parameters (drive amplitude, temperature, magnetic field).

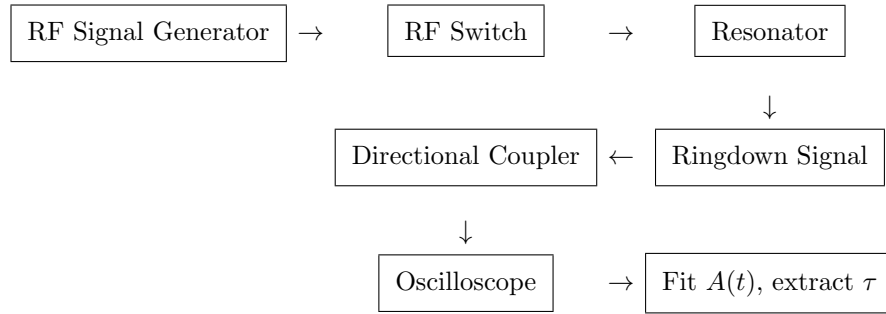


FIG. 1. Minimal tabletop resonator measurement architecture used to perform ringdown-based detection of deviations from standard cavity decay.

The ringdown is modeled as:

$$A(t) = A_0 e^{-t/\tau} \quad (1)$$

A deviation is defined as:

$$\Delta\tau = \tau_{\text{measured}} - \tau_{\text{expected}} \quad (2)$$

V. CONTROL TESTS

To exclude conventional artifacts, the following controls are required:

- dummy load with no cavity;
- detuned cavity measurement;

- zero magnetic field condition;
- repeated measurement under identical conditions;
- cable and instrumentation stability verification.

Any candidate signal must disappear or change predictably under these controls. These controls are intended to establish a null baseline and ensure that any observed deviations cannot be attributed to conventional electromagnetic, thermal, or measurement-system effects.

VI. EXPECTED SENSITIVITY AND DETECTABILITY

For a resonator with quality factor Q and angular frequency ω ,

$$\tau \approx \frac{Q}{\omega}. \quad (3)$$

The experiment is sensitive to fractional deviations:

$$\frac{\Delta\tau}{\tau} \ll 1. \quad (4)$$

For typical tabletop instrumentation, resolving fractional deviations on the order of 10^{-3} to 10^{-5} represents a practical lower bound, depending on noise stability and averaging.

This corresponds to absolute timing resolutions on the order of nanoseconds to microseconds, depending on resonator frequency and quality factor.

Given modest instrumentation, detectable deviations are expected to be near the measurement noise floor.

VII. INTERPRETATION

A null result implies:

- no detectable EVRT-like response at accessible scales;
- constraints on effective coupling strength;
- justification for higher-sensitivity experiments if warranted.

A positive result requires:

- reproducibility;
- parameter scaling behavior;
- independent replication;
- consistency under all control conditions.

VIII. LIMITATIONS

This experiment does not:

- provide microscopic interpretation;
- resolve weak signals below the noise floor;
- replace high-Q or cryogenic experiments.

It is intended solely as a preliminary screening test.

IX. CONCLUSION

A minimal tabletop resonator experiment provides a practical entry point for testing EVRT-like response signatures. The primary value of this approach lies in its accessibility: it enables rapid, low-cost testing before committing to high-precision infrastructure. A null result provides immediate constraints on the accessible parameter space, while any reproducible deviation would motivate higher-sensitivity experimental investigation.

The simplicity of the proposed setup allows rapid iteration and independent replication, making it suitable as a first-pass screening tool.